Education

KTH Royal Institute of Technology, Machine Learning — Sweden, 2023 – 2026

- *Thesis:* AI for Networked Robotics
- *Courses:* Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, Robotics, Computer Graphics, Multi-Agent Systems, Statistical Methods

University of Surrey, Computer Science — UK, 2019 – 2023

- *Thesis:* RoboKinesis
- Controlling Robotic Arm with Computer Vision
- *Courses:* Programming, Algorithms, Software and Web Engineering, Artificial Intelligence, Deep Learning, Natural Language Processing

Experience

Computational Imaging and Vision Laboratory, National Institute of Informatics, Research Exchange Student — Japan, 2025 – 2026, 1 year
Video Motion Magnification

- Analyzed reflectance behavior of synthetic materials using BRDF parameters and Video Motion Magnification techniques, producing promising results that will inform future lab research. Developed Python implementations of signal processing algorithms, including Riesz Transform for motion/reflectance decomposition.
- Supervisor: Dr. Imari Sato

Ericsson Research, Master's Thesis Student — Sweden, 2025 – 2025, 1 year
Towards Adaptive Reinforcement Learning for Network-Aware Robotics via Quantization Techniques

- Applied adaptive RL with quantization to network-aware robot control, demonstrating improved model efficiency and policy interpretability on proprietary datasets. Extended the base algorithm with stability improvements and real-world control constraint alignment. Built end-to-end training and evaluation pipelines for network-constrained environments.
- Supervisor: Dr. Christos N. Mavridis

Visualization Course, KTH Royal Institute of Technology, Teachning Assistant (Visualization course) — Sweden, 2024 – 2024, 1 year

- Graded coursework and led support sessions, guiding students through core visualization concepts.

Centre for Vision, Speech and Signal Processing, University of Surrey, Summer Research Internship Student — UK, 2022 – 2022, 1 year

- Built a Gazebo simulation environment and a curriculum-based Reinforcement Learning pipeline using PyTorch. Trained a TurtleBot agent to autonomously score goals in simulation, supporting future deployment for RoboCup robotics competitions.
- Supervisor: Dr. Oscar Mendez Maldonado

Hawk-Eye Innovations, Machine Learning Engineer (Placement) — UK, 2021 – 2022, 1 year

- Improved UX and functionality of an internal React-based annotation tool used for training models. Developed C++ applications for data collection, visualization, and preprocessing used by tennis operators. Optimized large-scale data processing pipelines, significantly reducing preparation time for tennis model training. Prototyped ML pipeline improvements during hackathons, including evaluating YOLO-based pose-estimation models in PyTorch to explore potential accuracy and latency gains.

Publications

Genetically Modified Wolf Optimization with Stochastic Gradient Descent for Optimising Deep Neural Networks
Novel optimization approach combining metaheuristic algorithms and SGD to balance exploration and exploitation for training Deep Neural Networks.
Manuel Bradicic, Michal Sitarz, Felix Sylvest Olesen
arxiv.org/abs/2301.08950

Skills

Programming Languages

ML/AI Frameworks

Tools & Infrastructure

3D/Simulation

Languages

Polish
Native speaker

English
Fluent

Serbo-Croatian
Fluent